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import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for numerical operations
import matplotlib.pyplot as plt # Import Matplotlib for plotting

# Step 1: Create a synthetic static background
bg = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array

# Step 2: Create a synthetic surveillance frame with a moving object (a square)
frame = bg.copy()
cv2.rectangle(frame, (40, 40), (60, 60), 255, -1) # Draw a filled or outlined rectangle

# Step 3: Initialize the Background Subtractor algorithm (Mid-Level CV feature)
backSub = cv2.createBackgroundSubtractorMOG2() # Create MOG2 background subtractor

# Step 4: Apply the algorithm to learn the background, then detect motion in the frame
backSub.apply(bg) # Apply frame to background subtractor to get foreground mask
motion_mask = backSub.apply(frame) # Apply frame to background subtractor to get motion mask

# Step 5: Visualize the extracted motion
plt.imshow(motion_mask, cmap='gray') # Render the image array to the display
plt.title('Mid-Level CV: Extracted Motion Mask') # Add a descriptive title to the plot
plt.show() # Show the final plot window to the user
```

